SMART INDIA HACKATHON 2026

- **Problem Statement ID** : SIH26105
- **Problem Statement Title** : AI-Powered Continuous Cyber Risk Quantification and Investment Optimization Platform
- **Theme** : Blockchain & CyberSecurity
- **PS Category** : Software
- **Team ID** : -
- **Team Name** : Powerhouse

PROPOSED SOLUTION

Continuous Risk Quantification

We convert every vulnerability into a financial number - ₹ Expected Annual Loss (EAL)

FAIR-Based Financial Engine

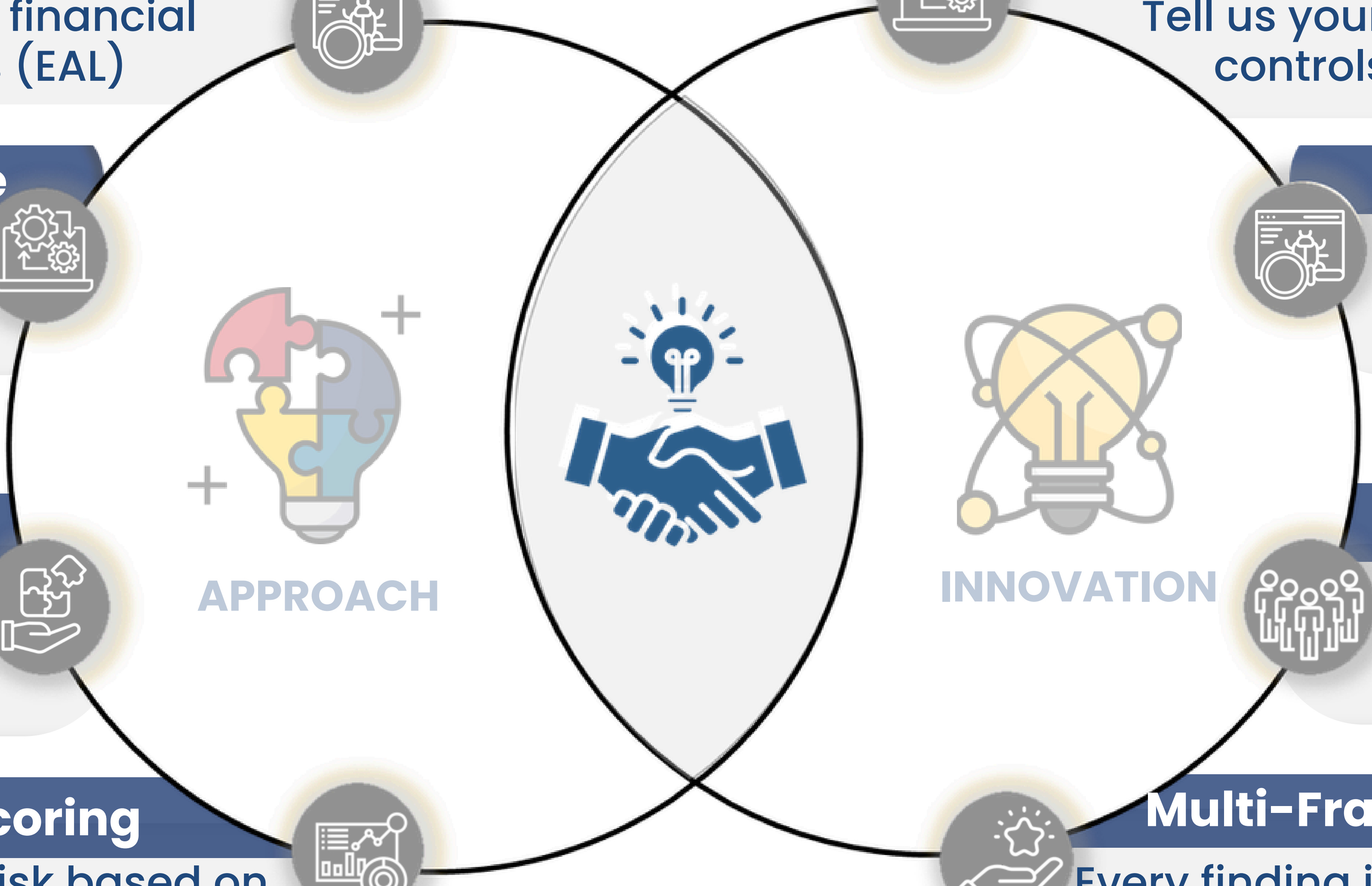
We use FAIR model to calculate exactly how much money your organization could lose from each threat.

India Regulatory Penalty Modeler

Every risk score includes real Indian RBI fines, SEBI charges and DPDP Act breach costs - 100% relevant to Indian

Asset-Weighted Risk Scoring

Not all servers are equal. We score risk based on how critical each asset is to the business, critical get higher risk than low-value assets.



AI Investment Optimizer

Tell us your ₹ budget, we tell you secutity controls that reduce the most risk.

Natural Language Query

Ask "What is our biggest cyber risk today?" - get answer with a ₹ figure. No technical knowledge needed.

What-If Scenario Simulator

if delay patching, CRISPR shows the exact ₹ increase in your financial exposure - before the damage happens.

Multi-Framework Compliance

Every finding is mapped to RBI CSF, SEBI, ISO 27001, NIST, and CIS - generating audit- generating audit-ready reports in one click.

From Raw Vulnerability Data to Board ₹ Risk Decisions In Real Time

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TECHNICAL APPROACH

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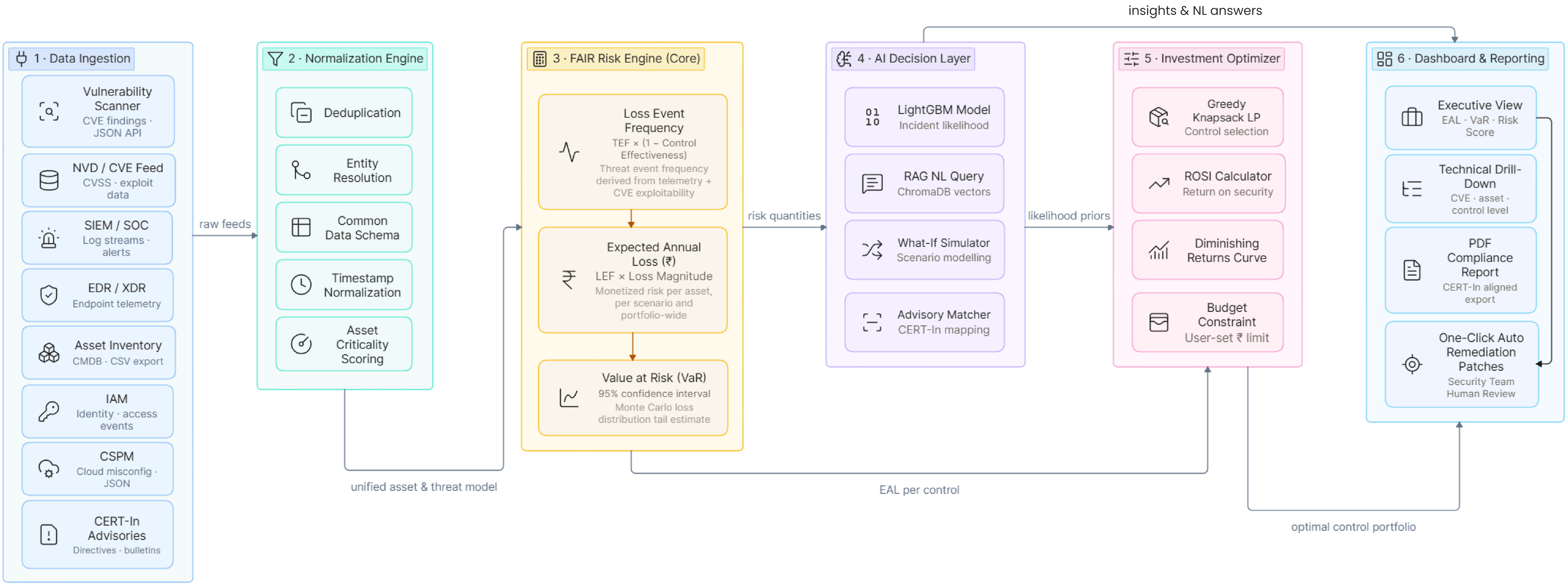
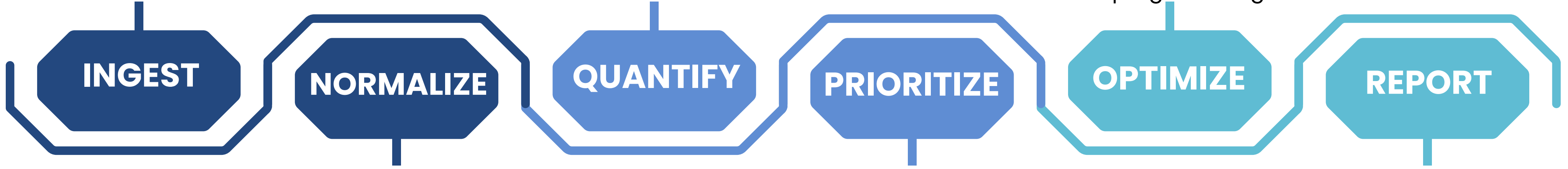


Security tools send data: vulnerability scanners, SIEM, IAM, EDR, CSPM, asset inventory

FAIR engine calculates EAL and VaR in ₹ for every asset and vulnerability combination

Investment optimizer finds the best set of controls for a given ₹ budget using linear programming

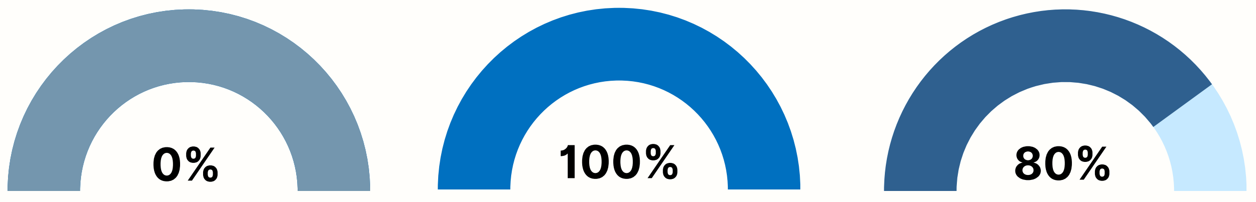


All data is cleaned, deduplicated, and mapped to a common schema for accurate comparison

AI ranks risks by financial impact — not CVSS score — so teams fix what costs the most first

Compliance-mapped PDF reports, executive dashboards, and NL query answers generated instantly

SECURITY GUARANTEES



DATA EGRESS

100% ₹-Denominated Output

REMEDIATION PATCHES

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FEASIBILITY AND VIABILITY

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FEASIBILITY

Built on the Proven FAIR Standard – **FAIR** is a recognized financial cyber-risk methodology. **CRISPR** implements the **calculation** in Python with **auditable logic**.

No External Dependencies for Core Function- Risk calculation and optimization can operate on **local** data without paid cloud subscriptions or proprietary **APIs**.

India Regulatory Data Built In- **RBI**, **SEBI**, **CERT**-In and **DPDP** Act context is incorporated into the financial impact model for Indian enterprises.

Modular and Scalable Architecture - Risk engine, AI layer, optimizer and dashboard are **separate** components that can be expanded through configuration.

RISK & CHALLENGES

Data Quality from Heterogeneous Sources- Enterprises use different tools and formats, making **one consistent risk** view difficult.

Accuracy of Financial Impact Estimates - Breachcost estimates can be **uncertain** without organization-specific historical **incident** data.

Adoption by Non-Technical Decision Makers - Boards may question **AI**-**generated** financial numbers if the methodology is not transparent.

STRATEGY & SOLUTIONS

Phased Rollout — Start Small, Prove Value Fast Begin with a small asset/vulnerability set, demonstrate live ₹ risk, then expand to enterprise scale.

Framework Validation by Regulatory Alignment - Cross-reference **calculations** with **RBI** and **NIST** controls to strengthen technical and audit confidence.

Continuous Learning from New CVE Data- **NVD** data and **CERT**-In advisories keep vulnerability and threat information current.

IMPACT AND BENEFITS

IMPACT

Towards a cyber-resilient India - where every rupee spent on security delivers measurable, provable protection for enterprises, banks, and government institutions.

Real-Time Risk Visibility	₹-Denominated Decisions	Regulatory Compliance	Budget Optimization	Nation-Scale Readiness

NATION-SCALE DEPLOYMENT

One platform built for startups, enterprises, banks and defense organizations.Adapts across cloud,on-premise and **air-gapped environments** with ease.

FINANCIAL IMPACT CLARITY

Indian organizations lose an average of **₹17.6** crore per data breach (IBM 2024). CRISPR makes this **risk visible** - before the breach happens. Boards can now answer: "**How much are we at risk, and what will fix it?**"

REDUCED REMEDIATION TIME

By prioritizing vulnerabilities by financial impact instead of CVSS score alone, security teams **fix the right things first -** reducing mean time to remediate critical risks by up to **60%**.

BENEFITS

Verified, Trustworthy Fixes - Every patch is sandbox-tested and confidence-scored before a human reviews it.

For the CISO - Evidence-Based Decisions

Stop presenting traffic-light dashboards to the board. Present ₹2.3 crore at risk, **reducible to ₹80 lakh with ₹50 lakh investment**. ROSI: 2.9x. Every budget conversation becomes a **data conversation**.

For the Board and Regulators - Instant Audit Readiness

Every risk finding is pre-mapped to **RBI CSF, SEBI CSCRF, ISO 27001, and DPDP Act 2023.** Compliance reports generate in **one click** - reducing audit preparation from weeks to hours.

For India - Closing Communication Gap

Today, 78% of Indian enterprises cannot quantify their cyber risk in financial terms (DSCI 2023). CRISPR closes that gap - giving **Indian organizations the same financial risk language that global enterprises use**, built specifically for Indian regulations and the Indian rupee.

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RESEARCH & REFERENCES

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Comparison Matrix

FEATURES	CRISPR	Bitsight (SaaS)	Tenable (SaaS)	Manual Assessment
₹-Denominated EAL Output	✓	✗	✗	✗
On-Premise / Zero Egress	✓	✗	✗	✓
RBI + SEBI Framework Mapping	✓	✗	✗	✗
AI Investment Optimizer	✓	✗	✗	✗
DPDP Act Penalty Estimator	✓	✗	✗	✗
NL Query Interface	✓	✗	✗	✗
What-If Scenario Simulator	✓	✓	✗	✗
Real-Time Continuous Update	✓	✓	✓	✗
Open Source · No Subscription	✓	✗	✗	✓

[1] Hubbard, D. W., & Seiersen, R. - "How to Measure Anything in Cybersecurity Risk", Wiley, 2023. Foundation of the FAIR methodology used in CRISPR's EAL calculator.

[2] IBM Security - "Cost of a Data Breach Report 2024" - Average breach cost in India: ₹17.6 crore. Used to calibrate CRISPR's loss magnitude estimates.

[3] DSCI (Data Security Council of India) - "India Cyber Threat Report 2023" - 78% of Indian enterprises cannot quantify cyber risk in financial terms. The exact problem CRISPR solves

AI & Tools We Use

FAIR Risk Model - "AI reasoning that never leaves the building." Drafts fixes and explanations fully inside your network.

LightGBM + ChromaDB - "ML that learns which vulnerabilities cost the most." LightGBM classifies incident likelihood using CVSS, patch age, and control effectiveness.

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NVD API + CERT-In Advisory Integration -"India's threat intel, wired directly into the risk engine." The National Vulnerability Database feeds live CVE data. CERT-In advisories trigger elevated risk scores for vulnerabilities actively targeting Indian enterprises.

FastAPI + React + Docker Compose - "Production-grade stack that runs on a single command." docker-compose up launches the full platform backend, AI layer, dashboard, and Nginx proxy in under 3 minutes.